JavaScript Foundation

Part 1 — Introduce JavaScript

HTML → Structure
CSS → Presentation
JavaScript → Behaviour

For example:

```
HTML
↓
Creates Login Form

CSS
↓
Makes Login Form Attractive

JavaScript
↓
Checks Username & Password
↓
Displays Error/Success
↓
Stores User Information
```

example

Create:

javascript-session1

- index.html
- script.js

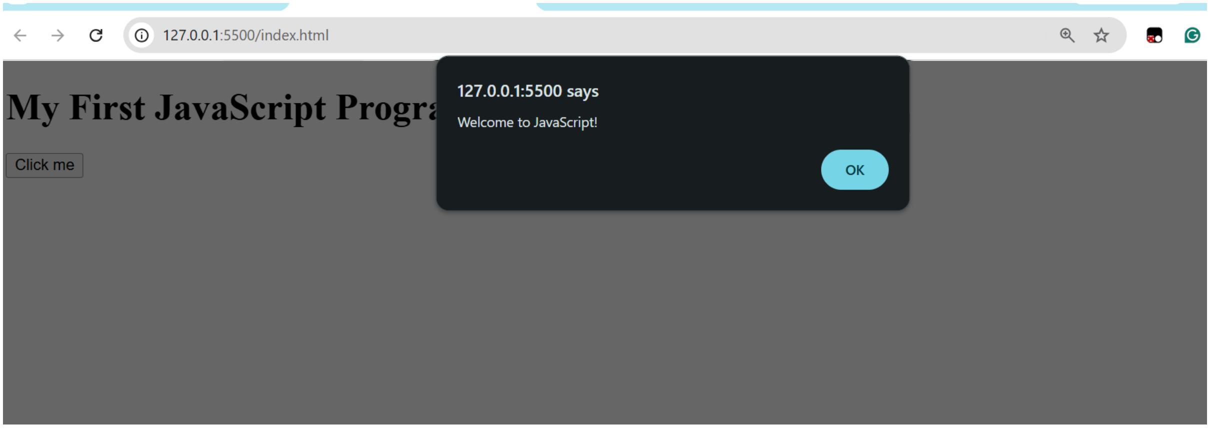
index.html

```html
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Introduction</title>
</head>
<body>

  <h1>My First JavaScript Program</h1>

  <button onclick="showMessage()">Click Me</button>

  <script src="script.js"></script>
</body>
</html>
```

script.js

```
function showMessage() {
  alert("Welcome to JavaScript!");
}
```

Output:



JavaScript can:

- respond to user actions
- change HTML
- validate forms
- store data
- communicate with servers
- create interactive websites

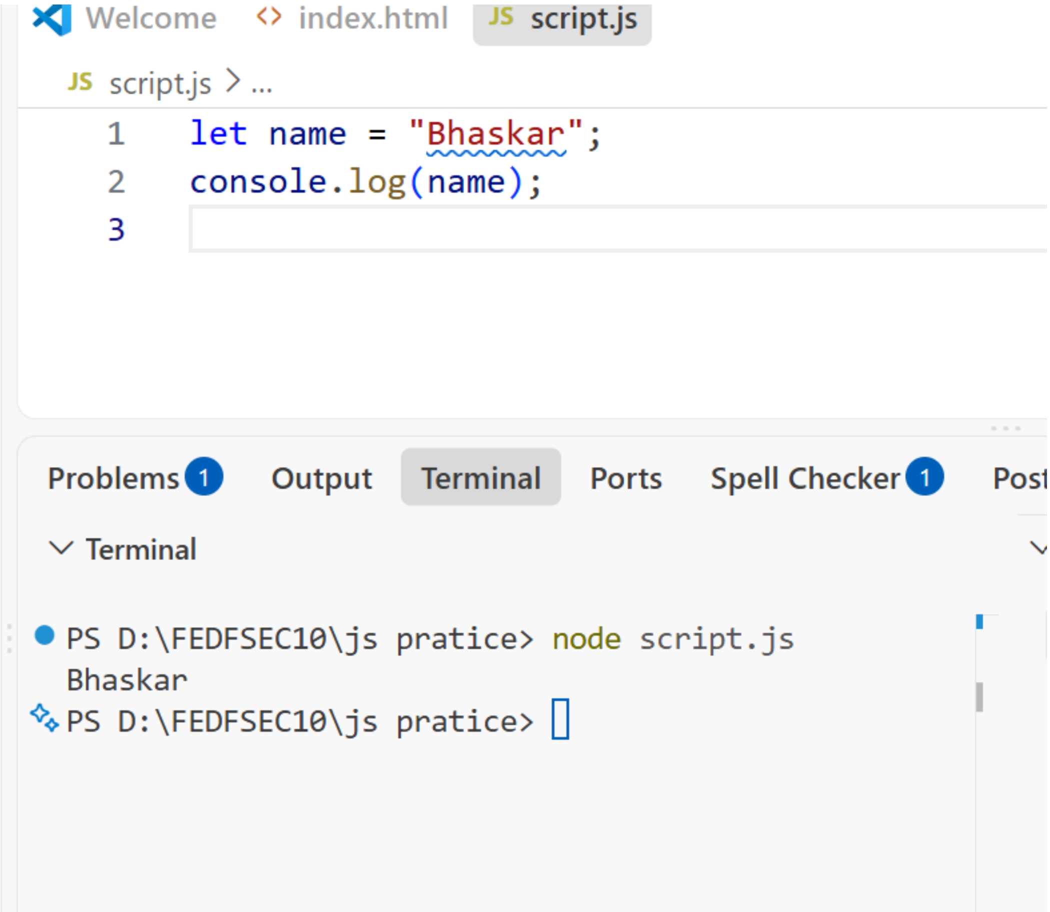
Part 2 — JavaScript Syntax

the basic structure:

```
let name = "Bhaskar";
console.log(name);
```

Description:

let → variable
name → variable name
= → assignment
"Bhaskar" → value
; → statement termination



Show comments:

```
// This is a single-line comment
```

and:

```
/*
  This is
  a multi-line comment
*/
```

Part 3 — Variables

```
let username = "student";
let age = 20;
const college = "KL University";
```

Description:

let

Value can change.

let age = 20;

age = 21;

const

Value should not be reassigned.

const college = "KL University";

example:

For our registration form, we will use variables to store username, email, password, etc.

Part 4 — Data Types

simple example:

```
let name = "Ravi";      // String
let age = 20;           // Number
let isStudent = true;   // Boolean
let marks;              // Undefined
let value = null;       // Null
console.log(typeof name);
console.log(typeof age);
console.log(typeof isStudent);
```

Expected:

string

number

boolean

Part 5 — Operators

Arithmetic

```
let a = 10;
let b = 5;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
```

Comparison

```
console.log(10 > 5);
console.log(10 == 10);
console.log(10 != 5);
```

Logical

```
let age = 20;
let hasID = true;
```

```
console.log(age >= 18 && hasID);
```

Description:

```
&& → AND
|| → OR
!  → NOT
```

These will be important for **form validation**.

Part 6 — Functions

```
function greet() {
   console.log("Welcome!");
}

greet();
```

```
function greetUser(name) {
   console.log("Welcome " + name);
}

greetUser("Ravi");
```

```
function add(a, b) {
   return a + b;
}

let result = add(10, 20);

console.log(result);
```

Explain:

```
Function
  ↓
Input
  ↓
Processing
  ↓
Output
```

Part 7 — Control Structures

if

```
let age = 20;

if (age >= 18) {
   console.log("Eligible");
}
```

if...else

```
let age = 16;

if (age >= 18) {
  console.log("Eligible");
} else {
  console.log("Not Eligible");
}
```

Multiple conditions

```
let marks = 75;

if (marks >= 90) {
  console.log("A");
} else if (marks >= 70) {
  console.log("B");
} else {
  console.log("C");
}
```

Description:

Form validation mainly uses if...else.

For example:

```
if (password.length < 6) {
  console.log("Password is too short");
}
```

Part 8 — DOM Manipulation

Create:

```
<h2 id="message">Hello</h2>

<button onclick="changeMessage()">
  Change Message
</button>
```

JavaScript:

```
function changeMessage() {

  document.getElementById("message").innerText =
    "Welcome to E-Commerce Website";
}
```

Explain:

```
HTML Element
   ↓
document.getElementById()
   ↓
JavaScript
   ↓
Change HTML
```

Reading input from HTML

Create:

```html
<input type="text" id="username">

<button onclick="displayName()">
  Submit
</button>

<p id="result"></p>
```

JavaScript:

```javascript
function displayName() {

  let username =
    document.getElementById("username").value;

  document.getElementById("result").innerText =
    "Welcome " + username;
}
```

Important concept

document.getElementById("username").value

means:

Get the value entered by the user in the input box.

Part 9 — Event Handling

An event is something that happens in the webpage.

Examples:

```
click
submit
change
input
mouseover
keydown
```

Instead of relying only on:

onclick=""

we can use:

```
document.getElementById("btn")
  .addEventListener("click", function() {

    alert("Button clicked");

  });
```

For forms:

```
document.getElementById("myForm")
  .addEventListener("submit", function(event) {

    event.preventDefault();

    console.log("Form submitted");

  });
```

Description for preventDefault():

Normally, submitting a form reloads the page. preventDefault() stops that default behaviour so JavaScript can validate the form.